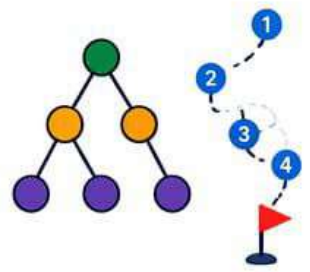




DATA STRUCTURES & ALGORITHMS (DSA) ROADMAP



STEP BY STEP GUIDE TO BECOME DSA PRO

LEARN → PRACTICE → MASTER → SUCCEED

1
2
3
4
5
6
7
8
9
10
11
12



1. Programming Basics

- Choose a Language (C++ / Java / Python)
- Basics: Variables, Loops, Functions, I/O
- Time & Space Complexity – Big O Notation

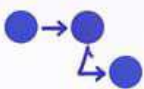
Duration
⌚ 1–2 Weeks
Resources
NCERT, W3Schools, Docs



2. Arrays & Strings

- Arrays, 2D Arrays
- String Basics
- Prefix Sum, Hashing Basics

Duration
⌚ 1–2 Weeks
Resources
LeetCode Explore, GFG



3. Linked List

- Singly Linked List
- Doubly Linked List
- Fast & Slow Pointers

Duration
⌚ 1–2 Weeks
Resources
LeetCode, GFG



4. Stack & Queue

- Stack (Implementation & Problems)
- Queue, Circular Queue, Deque
- Monotonic Stack/Queue

Duration
⌚ 1–2 Weeks
Resources
LeetCode, GFG



5. Trees

- Binary Tree Basics
- Tree Traversals (Inorder, Preorder, Postorder)
- BST (Insert, Search, Delete)

Duration
⌚ 2–3 Weeks
Resources
LeetCode, GFG, Visualgo



6. Binary Search Tree & Heaps

- BST Advanced Problems
- Heap (Min/Max Heap)
- Priority Queue

Duration
⌚ 1–2 Weeks
Resources
LeetCode, GFG



7. Graphs

- Graph Representations (Adj List, Matrix)
- BFS, DFS
- Topological Sort, Union Find (DSU)

Duration
⌚ 2–3 Weeks
Resources
LeetCode, GFG, VisuAlgo



8. Sorting & Searching

- Sorting: Bubble, Selection, Insertion
- Merge Sort, Quick Sort, Heap Sort
- Binary Search & Variations

Duration
⌚ 1–2 Weeks
Resources
LeetCode, GFG



9. Dynamic Programming

- Overlapping Subproblems
- Memoization vs Tabulation
- 0/1 Knapsack, LCS, LIS, Coin Change

Duration
⌚ 3–4 Weeks
Resources
LeetCode, GFG, DP Series



10. Recursion & Backtracking

- Recursion Basics
- N-Queens, Subsets, Permutations
- Sudoku Solver, Rat in a Maze

Duration
⌚ 2–3 Weeks
Resources
LeetCode, GFG



11. Advanced Topics

- Tries
- Segment Tree / Fenwick Tree (BIT)
- Greedy Algorithms, Interval Scheduling

Duration
⌚ 2–3 Weeks
Resources
LeetCode, GFG



12. Practice, Contests & Real Interviews

- Solve Problems Daily
- Participate in Contests
- Mock Interviews & Company Questions

Duration
⌚ Ongoing
Resources
LeetCode, Codeforces, AtCoder

CORE SKILLS YOU WILL BUILD

- 💡 Problem Solving
- 🧠 Logical Thinking
- 💻 Algorithm Design
- 🕒 Time & Space Optimization
- 🗄️ Data Handling Skills

IMPORTANT CONCEPTS

- Time Complexity
- Space Complexity
- Recursion
- Divide & Conquer
- Greedy
- Dynamic Programming
- Backtracking
- Graph Traversal
- Tree Traversal
- Hashing

BEST PLATFORMS TO PRACTICE



STUDY PLAN (SUGGESTED)

- ✅ Learn one topic at a time
- ✅ Solve basic problems
- ✅ Move to medium then hard
- ✅ Revise & read editorials
- ✅ Participate in contests
- ✅ Build consistency

CAREER OPPORTUNITIES

- 💻 Software Engineer
- 💼 SDE (Product Based Companies)
- 🏆 Competitive Programmer
- 📊 Data Engineer / Analyst
- 🤖 ML Engineer



Consistency + Practice + Patience = DSA Mastery

Solve. Learn. Improve. Repeat.

